Adding Variables to CR Plots, PE Plots, and/or Cox Analysis

This guide explains how the pipeline decides which variables are in the model and in the data, and how to add a variable (e.g. **div_name**, **yg**) to competing-risks (CR) plotting, partial-effects (PE) plots, and/or the Cox model.

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1. Static vs time-varying

- **Static** (e.g. **yg**, final_job_code): One value per officer for the whole follow-up. Cell 10 takes them from officer-level data and repeats that value for every interval. They belong in **static** config (`model_static_cols` or an extra-static list), **not** in time-varying lists.
- **Time-varying** (e.g. **div_name**): Can change from snapshot to snapshot (e.g. officer moves divisions). Cell 10 takes them from each snapshot, so each interval gets the value at that time. They belong in **time-varying** config (`EXTRA_TIME_VARYING_COLS` → `model_time_varying_cols`).

Impact: Static vars give "effect of being in group X" (constant for the officer). Time-varying vars give "effect of currently being in X" and have within-officer variation, which helps identify effects. Use the correct list so the pipeline builds the right structure.

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2. What actually puts a variable in the model and in the data

2.1 Single source of truth for "in the pipeline and in the model"

- **Time-varying:**
  `EXTRA_TIME_VARYING_COLS` (in your override, e.g. `pipeline_config_17_1.py`) is appended to `model_time_varying_cols`.
  Then `time_varying_cols` is set to:
  `base + oer + prestige + model_time_varying_cols`.
  So anything in `model_time_varying_cols` is automatically carried into Cell 10 (survival data → `df_cox`) and into the Cox model variable list in Cell 12. No need to add the same variable to a second list.

- **Static:**
  Use `model_static_cols` (or an **EXTRA_STATIC_COLS** list you append to it). That controls which static variables are in the model. Static columns must also be present in the data that feeds Cell 10 (e.g. in officer_info / the frame that has one row per officer).

2.2 Dummy creation: who creates dummies?

- `COLUMN_CONFIG['dummy_variables']['categorical_cols']` (in base `pipeline_config.py`):
  Cell 11 **only** creates dummies for columns that are (a) listed in `categorical_cols` and (b) present in `df_analysis`. So if a column isn't in the pipeline (e.g. you removed it from `EXTRA_TIME_VARYING_COLS`), Cell 11 skips it and it never enters the model.

- **Cell 12** has **hardcoded** logic for **yg**, **final_job_code**, and **div_name**. For those three, Cell 12 can create dummies itself (when `create_dummies` is True) if they're in the model list and still in the dataframe. So for div_name you can leave it out of `categorical_cols` and Cell 12 will still create div dummies; adding div_name to `categorical_cols` just means Cell 11 creates them instead.

2.3 "I don't want this variable to confound the model right now"

- **Remove it from the list that puts it in the model.**
  For time-varying: remove it from `EXTRA_TIME_VARYING_COLS`.
  For static: remove it from `model_static_cols` (or your EXTRA_STATIC_COLS).
- You can **leave** it in `categorical_cols`. When the variable isn't in the pipeline, it won't be in `df_cox` / `df_analysis`, so Cell 11 will skip it ("column not found"). It will not be in the model, so no confounding or extra collinearity.

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3. How to add a variable: step-by-step

3.1 Add a time-varying variable (e.g. div_name) to Cox analysis and/or plots

Goal: Variable is in the survival data, in the Cox model (as dummies), and available for CR/PE plotting.

1. **Create the column earlier in the pipeline** (e.g. Cell 7 for `div_name` when `DIVISION_CONFIG['enabled'] = True`). Ensure it exists on the snapshot frame that feeds Cell 10.
2. **In your override (e.g. `pipeline_config_17_1.py`):**
   - Add the column name to `EXTRA_TIME_VARYING_COLS`:

     ```
     EXTRA_TIME_VARYING_COLS = [
         'div_name',   # add others as needed
     ]
     COLUMN_CONFIG['model_time_varying_cols'] = list(COLUMN_CONFIG.get('model_time_varying_cols', [])) +
             EXTRA_TIME_VARYING_COLS
     ```

   - Ensure `create_dummies` is True if the variable is categorical:

     ```
     COLUMN_CONFIG['dummy_variables']['create_dummies'] = True
     ```

3. **Optional:** Add it to `categorical_cols` in base `pipeline_config.py` (uncomment or add an entry) if you want Cell 11 to create the dummies; otherwise Cell 12 will create them for div_name.
4. `time_varying_cols` is already built from `model_time_varying_cols` in the override, so the variable is carried into Cell 10 and Cell 11 automatically. No extra step.
5. **Re-run** from Cell 7 (if you added a new column) and Cell 10 onward so `df_cox` and the model include the variable.

CR plot with this variable (e.g. group by div_name):
Set `USE_OVERRIDE_CR_PLOTS = True` and add a plot that uses the variable (e.g. `group_by='div_name'`, and optionally `variable=...`, `n_bins=1`). The variable must be in `df_cox` (so it must be in `EXTRA_TIME_VARYING_COLS` / `model_time_varying_cols` as above).

PE plots:
Time-varying variables that are in the model will appear in the set of variables considered for partial-effects plots (Cell 12.6). Division dummies (e.g. `div_82ABN`) can appear as categorical bars in 12.6 if they're in the key-vars list; 12.6A is for combined linear+quadratic terms.

3.2 Add a static variable (e.g. yg) to Cox analysis and/or plots

Goal: Variable is in the Cox model (as dummies) and available for plotting, without putting it in time-varying lists.

1. **Ensure the column exists** in the data that feeds Cell 10 (e.g. officer-level frame with one row per officer; yg is usually already there from Cell 3 or 502).
2. **In your override:**
   - Append it to `model_static_cols` (or use an **EXTRA_STATIC_COLS** list):

     ```
     EXTRA_STATIC_COLS = ['yg']   # or [] when you don't want it
     COLUMN_CONFIG['model_static_cols'] = list(COLUMN_CONFIG.get('model_static_cols', [])) + EXTRA_STATIC_COLS
     ```

   - Ensure `create_dummies` is True if it's categorical.
   - **Do not** add it to `EXTRA_TIME_VARYING_COLS` (that's for time-varying only).
3. **Optional:** Add **yg** (or the column) to `categorical_cols` in base config if you want Cell 11 to create dummies; otherwise Cell 12 will create yg dummies via its hardcoded yg block.
4. **Re-run** from Cell 10 (or earlier if the static column is built earlier) so the static variable is in the model.

CR plot:
Use the variable as `group_by` in an override CR plot (e.g. `group_by='yg'`). It must be in the data and in the model for the plot to use it.

3.3 Add a variable only to CR plotting (not to the Cox model)

Goal: Stratify or group CR plots by a variable without including it in the Cox model (to avoid confounding).

1. **Get the variable into the pipeline data only** (so it's in `df_cox` for plotting), but **not** into `model_time_varying_cols` or `model_static_cols`.
   - For a **time-varying** column: you'd need it in `time_varying_cols` but **not** in `model_time_varying_cols`. Currently we set `time_varying_cols = base + oer + prestige + model_time_varying_cols`, so every model time-varying column is in the pipeline. To have a column in the pipeline but not in the model you'd add it to the base/oer/prestige part of `time_varying_cols` only (e.g. in base config or a separate "plot-only" list that gets merged into `time_varying_cols` but not into `model_time_varying_cols`). Implementation depends on your config layout.
   - For a **static** column: it would need to be in the officer-level data and in whatever column list Cell 10 uses for static columns, but not in `model_static_cols`.
2. **CR plot:** Set `USE_OVERRIDE_CR_PLOTS = True` and add a plot with the desired `variable` and/or `group_by`. The variable must exist in the data used for the CR plot (usually `df_cox` or the same analysis dataframe).

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4. Quick reference

Goal	Time-varying (e.g. div_name)	Static (e.g. yg)
In Cox model	Add to **EXTRA_TIME_VARYING_COLS** (in override)	Add to **model_static_cols** or **EXTRA_STATIC_COLS**
In pipeline data (Cell 10)	Automatic once in model_time_varying_cols	Must be in officer-level data and in static column list
Dummies	create_dummies=True; optional: add to categorical_cols	Same
CR plot by this variable	USE_OVERRIDE_CR_PLOTS=True, add plot with group_by='div_name' (and variable/n_bins as needed)	Same idea with group_by='yg'
Don't want in model	Remove from EXTRA_TIME_VARYING_COLS	Remove from model_static_cols / EXTRA_STATIC_COLS

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5. Legacy note (run overrides and CR defaults)

- **RUN_PROFILE** and **expand_run_profile()** set the main time-varying model columns (tv_vars, quadratics, interaction) and the default CR plot set. CR options (bins, etc.) go in **RUN_CR_PLOT_DEFAULTS** and `'cr_plot_defaults'` in RUN_PROFILE.
- To customize CR plots per plot (e.g. one plot with `n_bins=1`, `group_by='div_name'`), set **USE_OVERRIDE_CR_PLOTS = True** and define the full **PLOT_CONFIG['plots']** list in the override (see commented example in `pipeline_config_17_1.py`).